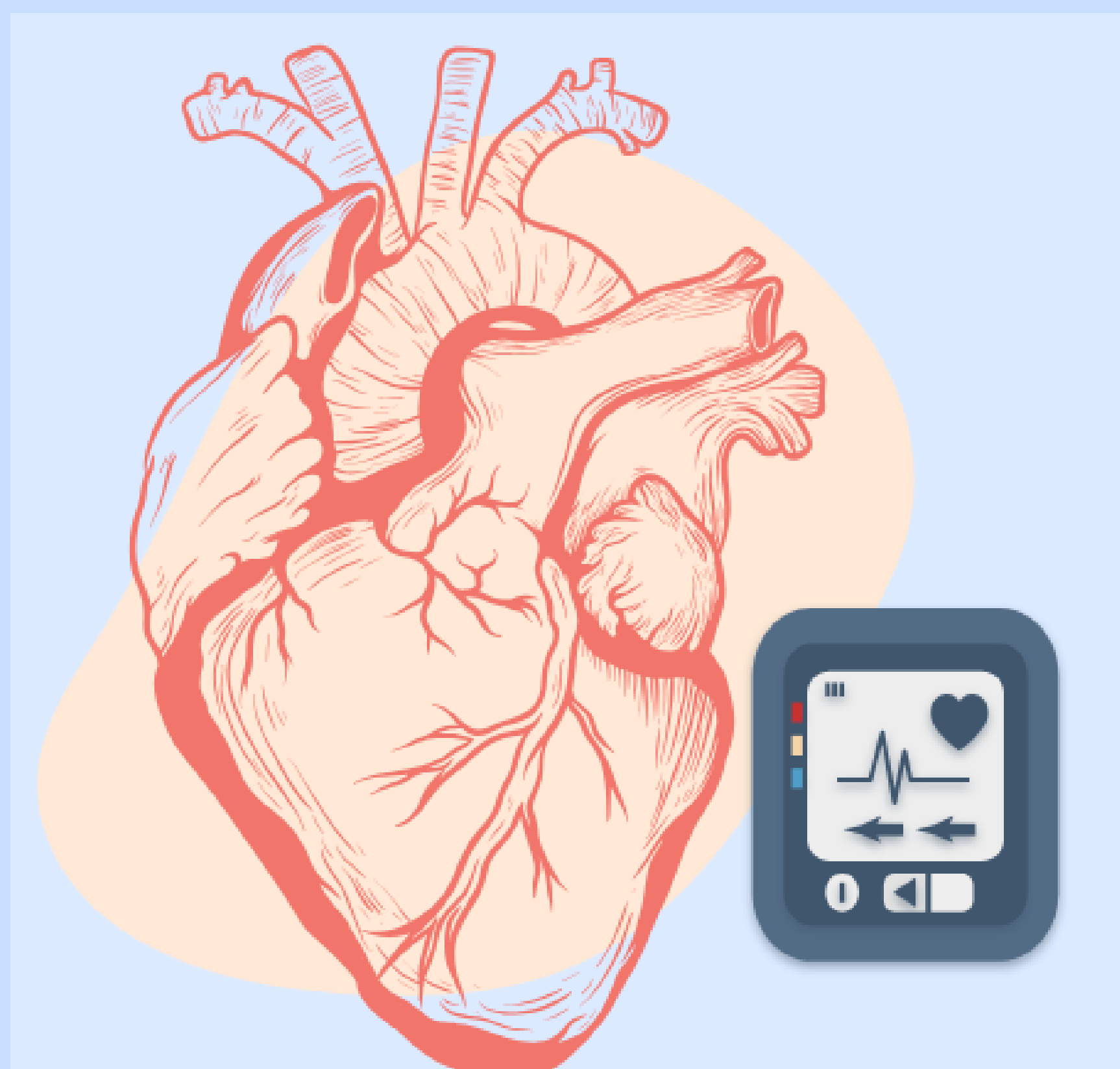


LifeFactors

Unveiling Disease Probability with Machine Learning

HEART DISEASE & DIABETES



Diabetes

Heart Disease

Data Sets

- An estimated **462 million individuals** are affected by type 2 diabetes (6.28% of the world's population)
- Cardiovascular diseases are the leading cause of death globally (estimated **17.9 million lives** each year)

1. **Diabetes** - BFRSS dataset, CDC, 2015
2. **Heart-Specific Blood** work specific dataset, 2018
3. **CHD** & Myocardial infarction (BFRSS Dataset, CDC 2020)

ACCURACY SCORES - ML MODEL

Random Forest Classifier

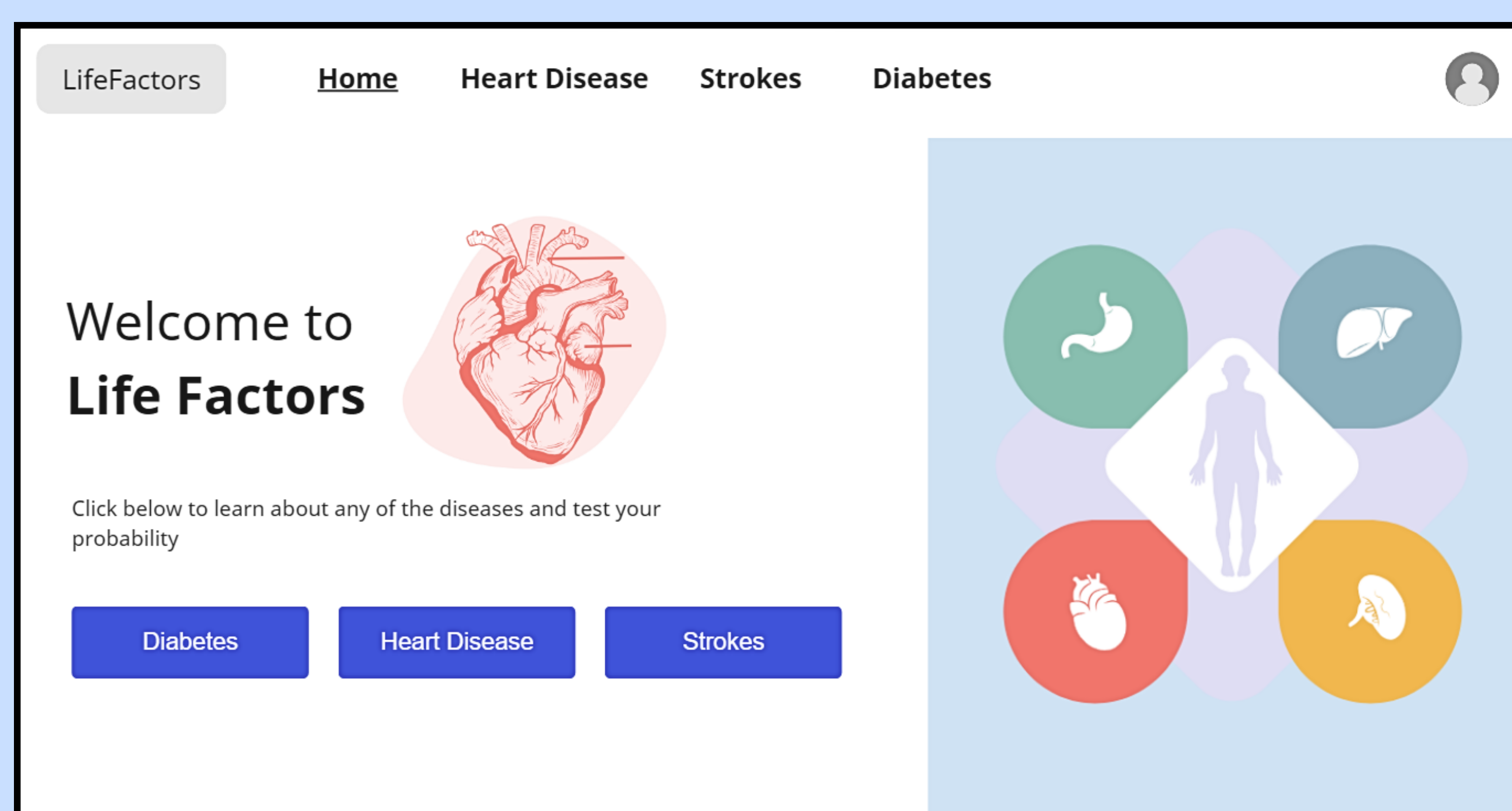
Diabetes 74.48 %
Heart-bloodwork 85.24 %
Heart 78.55 %

K-Nearest Neighbors

Diabetes 69.88 %
Heart-bloodwork 78.69 %
Heart 73.40 %

Gaussian Naïve Bayes

Diabetes 71.69 %
Heart-bloodwork 81.87 %
Heart 69.45 %



Support Vector Machine

Diabetes 75.33 %
Heart-Bloodwork 88.52 %
Heart 78.40 %

Logistic Regression

Diabetes 75.64 %
Heart-Bloodwork 78.69 %
Heart 78.66 %

Gradient Boosting Classifier

Diabetes 74.85 %
Heart-Bloodwork 86.89 %
Heart 78.20 %

Project Significance

- Personalized risk probability
- Lifestyle Targeted interventions
- Costs & Resources
- Invasive Medical Procedures: Biopsies, Angiographies, Endoscopic procedures

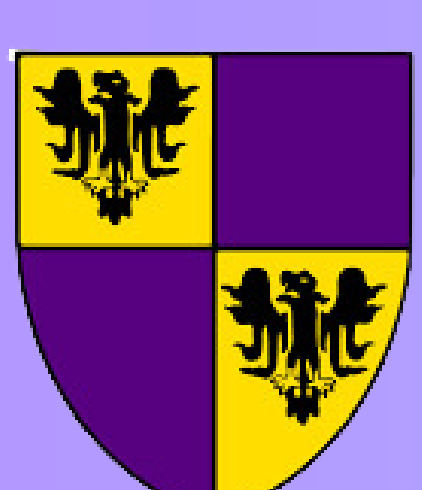
ABOUT

SEMESTER 2/2023

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